

Research Notes on Logistic regression

Logistic regression

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

$$P(y=1|x) = \sigma(w^T x + b)$$

>> Section 1

OR = odds ■ (x + 1) odds ■ (x) = (p (x + 1) 1 − p (x + 1)) (p (x) 1 − p (x)) = e β 0 + β 1 (x + 1) e β 0 + β 1 x = e β 1 { \displaystyle \mathrm {OR} = \frac {\operatorname {odds} \{x+1\}}{\operatorname {odds} \{x\}} = \frac {\left(\frac {p(x+1)}{1-p(x+1)}\right)}{\left(\frac {p(x)}{1-p(x)}\right)}} = \frac {e^{\beta _{0}+\beta _{1}(x+1)}}{e^{\beta _{0}+\beta _{1}x}} = e^{\beta _{1}} }

>> Section 2

Using the chi-squared test of significance, the integral of the chi-squared distribution with one degree of freedom from 11.6661... to infinity is equal to 0.00063649... This effectively means that about 6 out of a 10,000 fits to random yk can be expected to have a better fit (smaller deviance) than the given yk and so we can conclude that the inclusion of the x variable and data in the proposed model is a very significant improvement over the null model. In other words, we reject the null hypothesis with 1 − D ≈ 99.94 % { \displaystyle 1-D\approx 99.94\% } confidence.

>> Section 3

A graph of the logistic function on the t-interval (−6,6) is shown in Figure 1. Let us assume that t { \displaystyle t } is a linear function of a single explanatory variable x { \displaystyle x } (the case where t { \displaystyle t } is a linear combination of multiple explanatory variables is treated similarly). We can then express t { \displaystyle t } as follows:

>> Section 4

General Logistic regression is used in various fields, including machine learning, most medical fields, and social sciences. For example, the Trauma and Injury Severity Score (TRISS), which is widely used to predict mortality in injured patients, was originally developed by Boyd et al. using logistic regression. Many other medical scales used to assess severity of a patient have been developed using logistic regression. Logistic regression may be used to predict the risk of developing a given disease (e.g. diabetes; coronary heart disease), based on observed characteristics of the patient (age, sex, body mass index,

results of various blood tests, etc.). Another example might be to predict whether a Nepalese voter will vote Nepali Congress or Communist Party of Nepal or for any other party, based on age, income, sex, race, state of residence, votes in previous elections, etc. The technique can also be used in engineering, especially for predicting the probability of failure of a given process, system or product. It is also used in marketing applications such as prediction of a customer's propensity to purchase a product or halt a subscription, etc. In economics, it can be used to predict the likelihood of a person ending up in the labor force, and a business application would be to predict the likelihood of a homeowner defaulting on a mortgage. Conditional random fields, an extension of logistic regression to sequential data, are used in natural language processing. Disaster planners and engineers rely on these models to predict decisions taken by householders or building occupants in small-scale and large-scales evacuations, such as building fires, wildfires, hurricanes among others. These models help in the development of reliable disaster managing plans and safer design for the built environment.

>> Section 5

In terms of binomial data A closely related model assumes that each i is associated not with a single Bernoulli trial but with ni independent identically distributed trials, where the observation Yi is the number of successes observed (the sum of the individual Bernoulli-distributed random variables), and hence follows a binomial distribution:

>> Section 6

t = log b ■ p 1 − p = β 0 + β 1 x 1 + β 2 x 2 + ■ + β M x M { \displaystyle t=\log _{b}\frac {p}{1-p}=\beta _{0}+\beta _{1}x_{1}+\beta _{2}x_{2}+\cdots +\beta _{M}x_{M}} }

>> Section 7

which is maximized using optimization techniques such as gradient descent. Assuming the (x , y) { \displaystyle (x,y)} pairs are drawn uniformly from the underlying distribution, then in the limit of large N,

>> Section 8

Pr (Y i = c) = e β c ⋅ X i ∑ h e β h ⋅ X i { \displaystyle \Pr(Y_{i}=c)=\frac {e^{\boldsymbol {\beta }}_{c}}{\sum _{h}e^{\boldsymbol {\beta }}_{h}}\cdot \mathbf {X} _{i}} } }

>> Section 9

Although several statistical packages (e.g., SPSS, SAS) report the Wald statistic to assess the contribution of individual predictors, the Wald statistic has limitations. When the regression coefficient is large, the standard error of the regression coefficient also tends to be larger

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increasing the probability of Type-II error. The Wald statistic also tends to be biased when data are sparse.

>> Section 10

Y is the Bernoulli-distributed response variable and x is the predictor variable; the β values are the linear parameters. The logit of the probability of success is then fitted to the predictors. The predicted value of the logit is converted back into predicted odds, via the inverse of the natural logarithm – the exponential function. Thus, although the observed dependent variable in binary logistic regression is a 0-or-1 variable, the logistic regression estimates the odds, as a continuous variable, that the dependent variable is a 'success'. In some applications, the odds are all that is needed. In others, a specific yes-or-no prediction is needed for whether the dependent variable is or is not a 'success'; this categorical prediction can be based on the computed odds of success, with predicted odds above some chosen cutoff value being translated into a prediction of success.

>> Section 11

$$-D_{KL}(Y \parallel Y_{\theta}) = H(Y \parallel X) \quad \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N \log \Pr(y_i \mid x_i; \theta) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \Pr(X=x, Y=y) \log \Pr(Y=y \mid X=x; \theta)$$
$$= -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \Pr(X=x, Y=y) \left(-\log \frac{\Pr(Y=y \mid X=x)}{\Pr(Y=y \mid X=x) \Pr(Y=y)} \right) = -D_{KL}(Y \parallel Y_{\theta}) - H(Y \mid X)$$

>> Section 12

Parameter estimation Since η is nonlinear in β_0 and β_1 , determining their optimum values will require numerical methods. One method of maximizing η is to require the derivatives of η with respect to β_0 and β_1 to be zero: